

Differential Expression and Enrichment Analysis of Type 2 Diabetes Adipose Tissue

BCB420

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Background and Question

- Type 2 diabetes alters adipose tissue biology
- Adipose tissue regulates metabolism and insulin sensitivity
- Molecular mechanisms underlying diabetes
- Obesity is very common among people with diabetes
- **Research question: What genes and pathways are altered in T2D adipose tissue led to obesity?**

GEO
GSE104674

48 samples

Preprocessing and QC

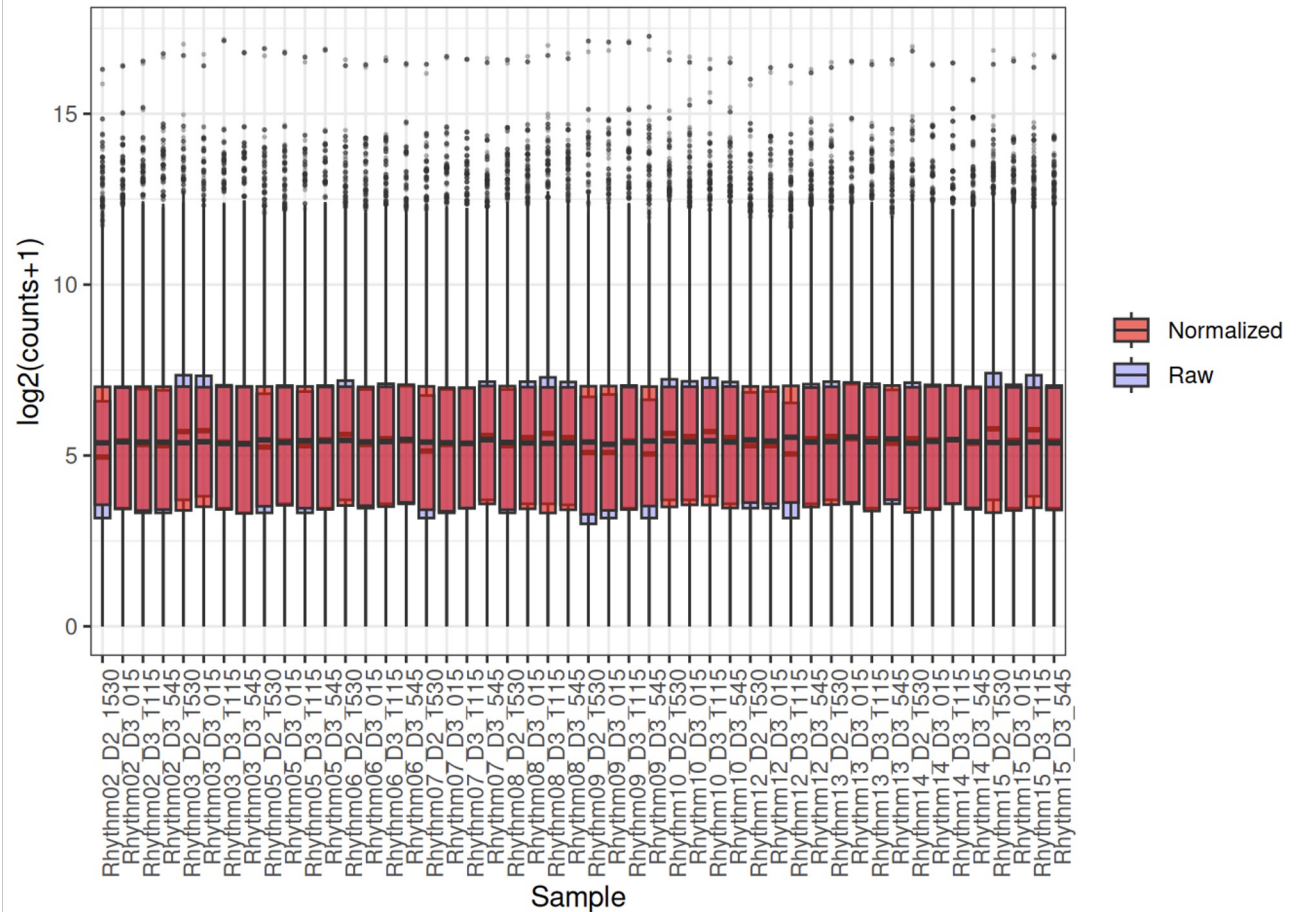
- Raw counts + metadata
- Low-expression filtering
- Gene-level summarization
- Gene symbol mapping
- Normalization

Summary: 9,029 genes

removed

15,335 genes × 48 samples

No major outliers



Differential Expression

DESeq2

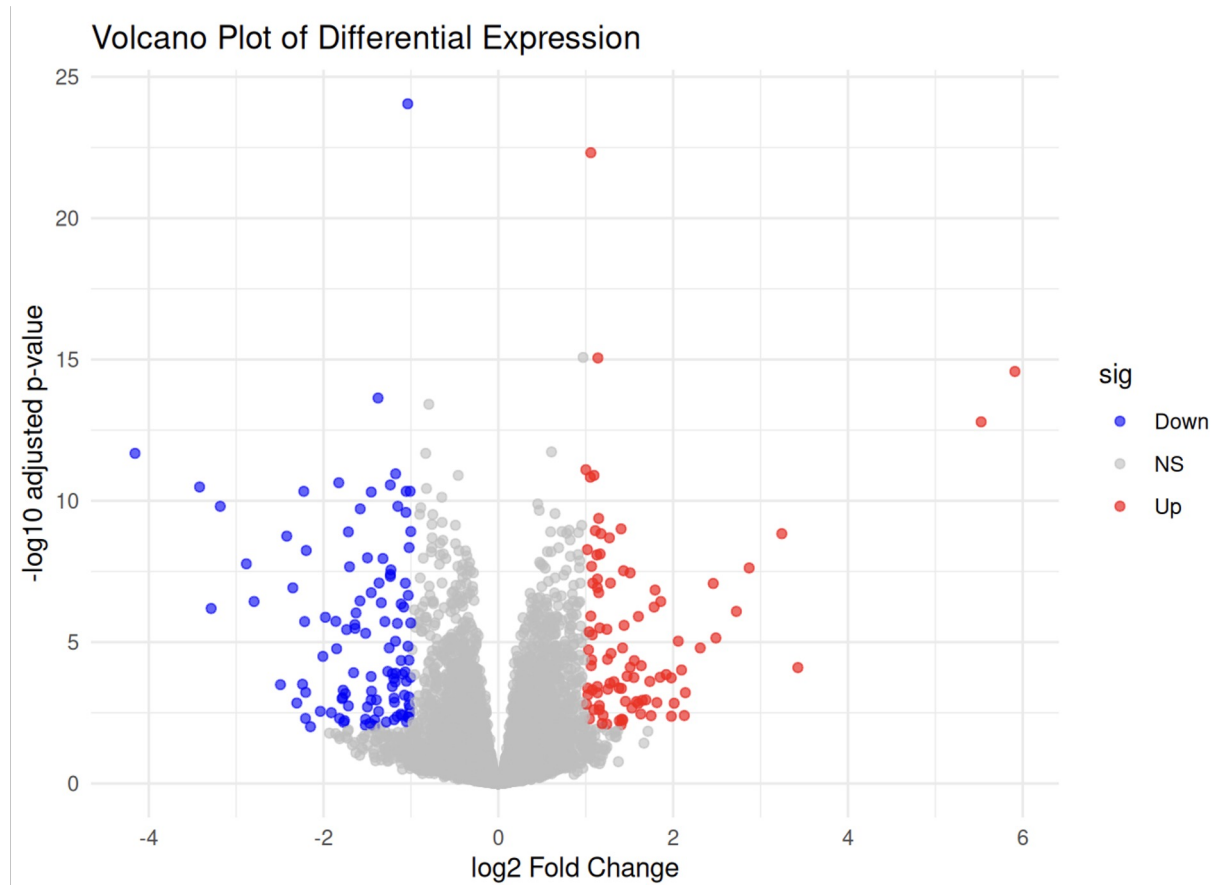
- Design: diabetes VS control
- BH correction

Cutoffs

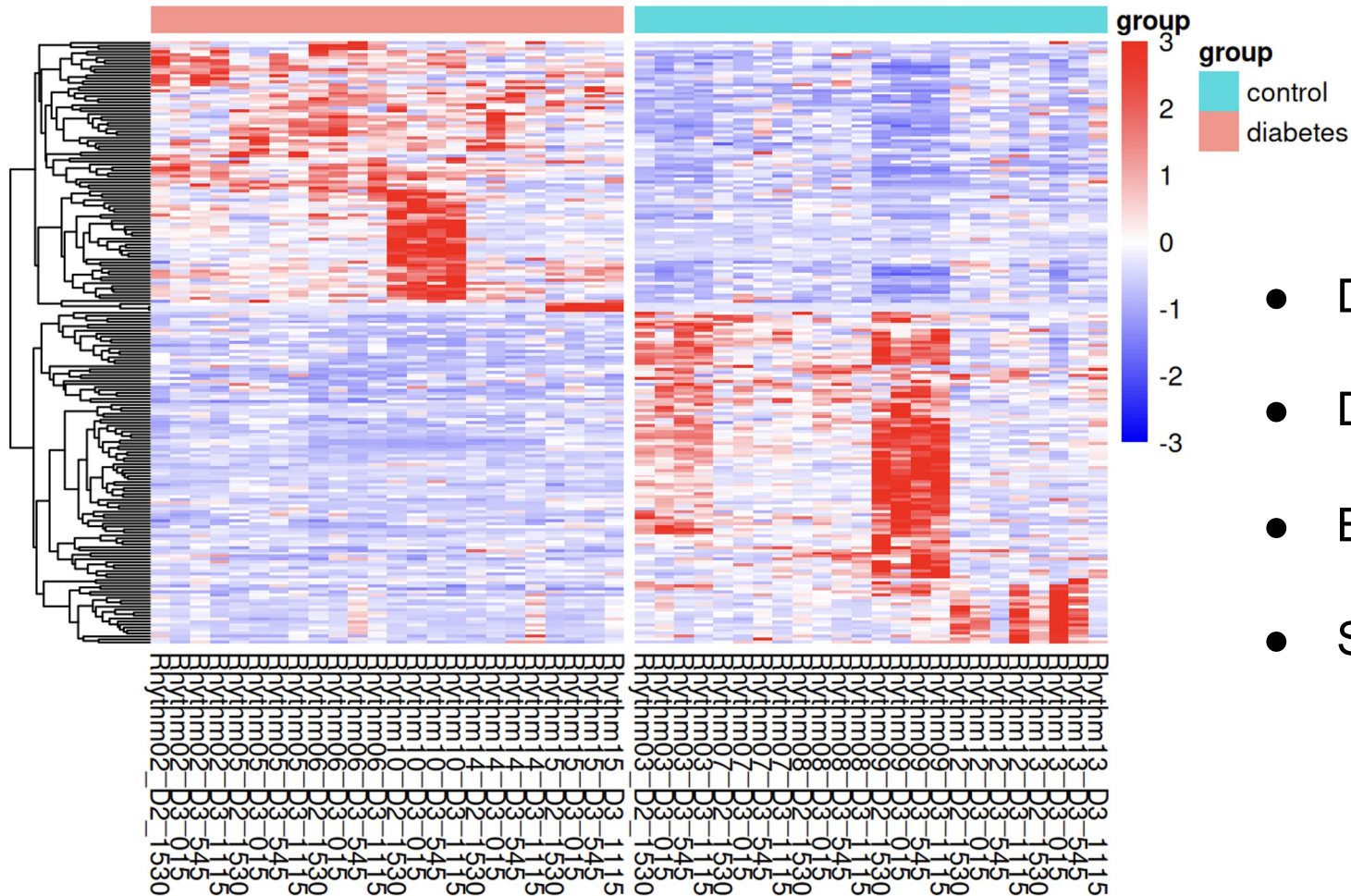
- Adjusted $p < 0.01$
- $|\log_2\text{FC}| > 1$

Result

- 203 significant DEGs
- 92 upregulated
- 111 downregulated



Heatmap of Differentially Expressed Genes

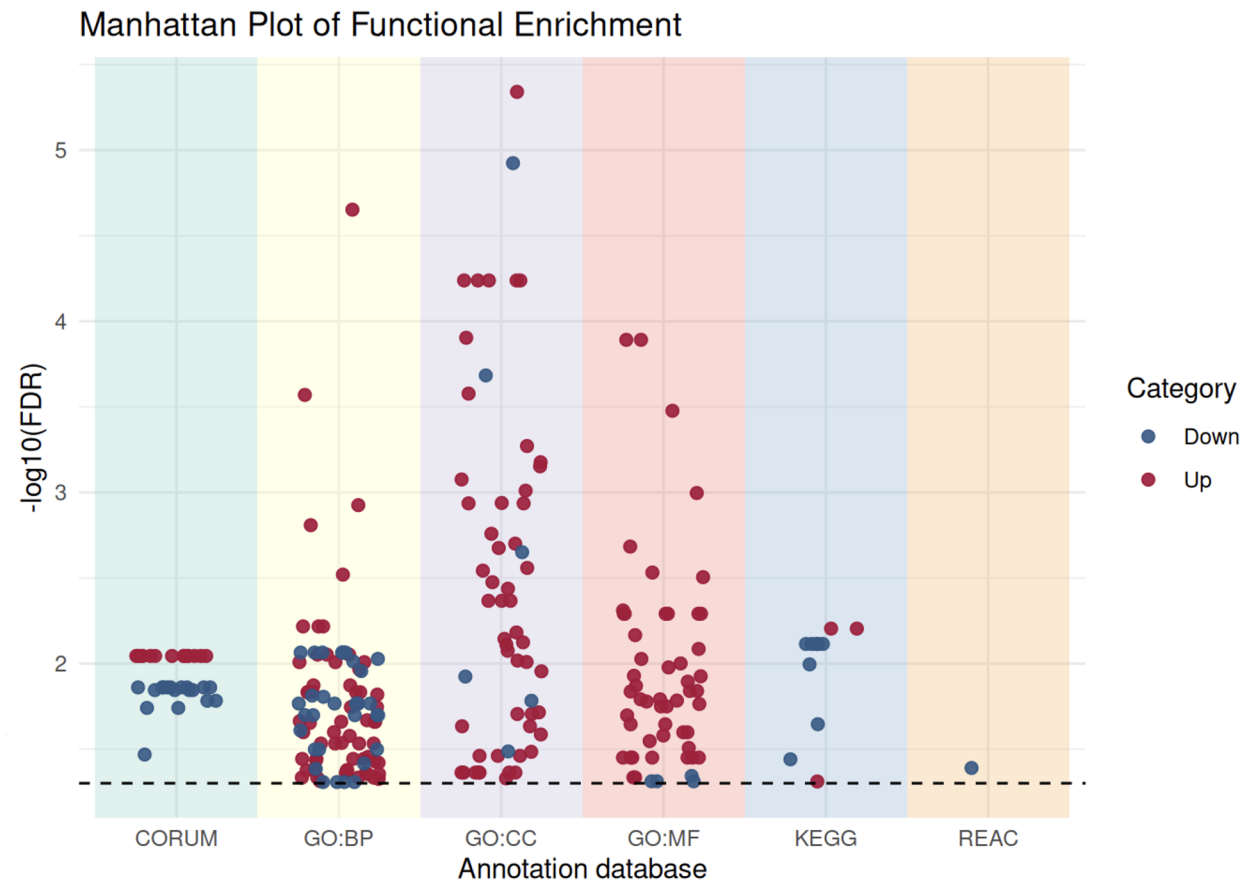


Expression Pattern Across Samples

- Differentially expressed genes
- Distinct heatmap blocks
- Both up and down
- Systematic transcriptomic

Over-representation analysis in g:Profiler

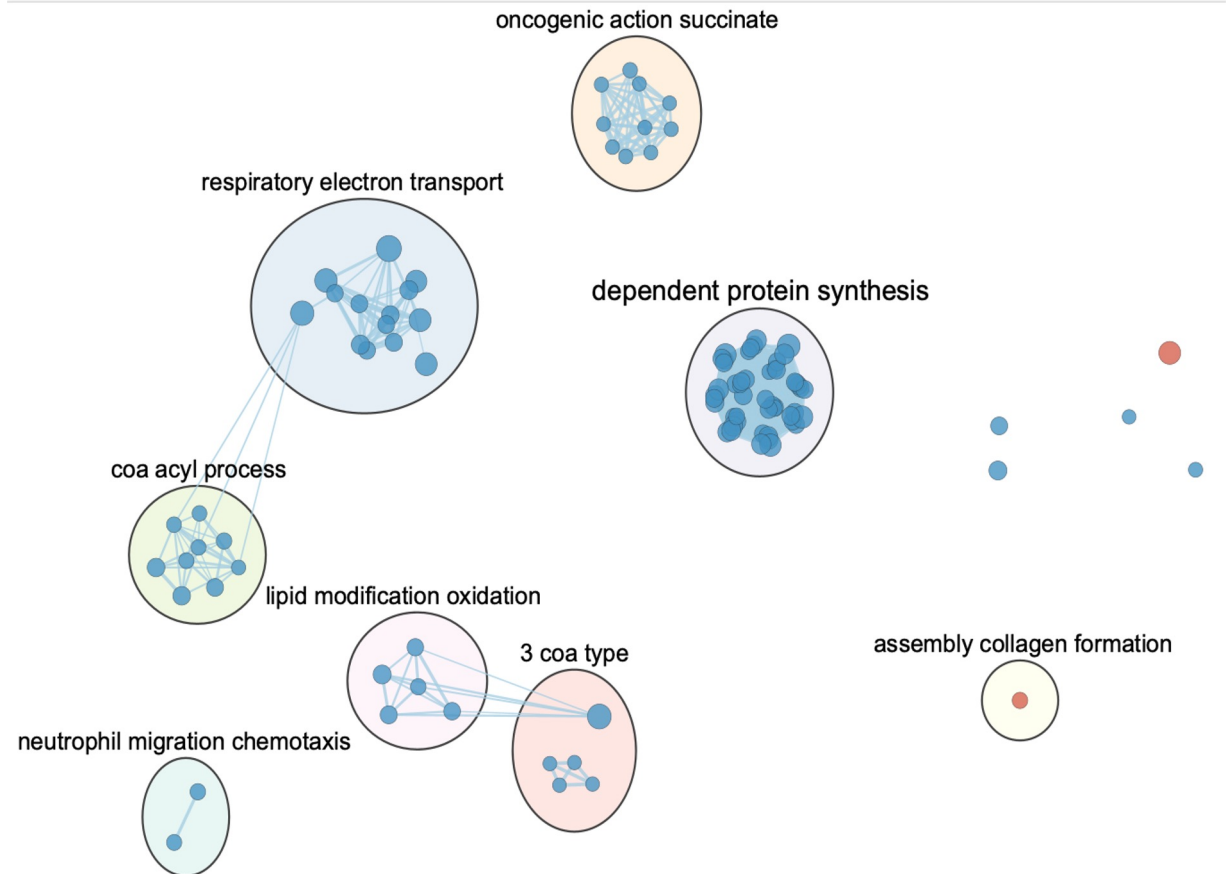
Database	Term	log10FDR	GeneCount	Direction
GO:CC	cell periphery	4.924	56	Down
GO:CC	plasma membrane	3.684	50	Down
GO:CC	CA3 pyramidal cell dendrite	2.651	2	Down
KEGG	Pyruvate metabolism	2.115	4	Down
KEGG	Nitrogen metabolism	2.115	3	Down
GO:CC	cell periphery	5.340	48	Up
GO:BP	multicellular organismal process	4.652	53	Up
GO:CC	extracellular space	4.239	30	Up
GO:CC	extracellular region	4.239	35	Up
GO:CC	synaptic membrane	4.239	11	Up



GSEA and Pathway Network

Cutoffs: FDR - 0.005, edge similarity - 0.5

Clusters (20)	...	Coll...
dependent protein synthesis	51	
3 coa type	25	
respiratory electron transport	21	
oncogenic action succinate	14	
coa acyl process	12	
lipid modification oxidation	7	
antigen mhc class	4	
assembly collagen formation	3	
regulation fatty acid	3	
cd8 tcr naive	2	
defects toll receptor	2	
defense response fungus	2	
ephrin reverse signaling	2	
neutrophil migration chemotaxis	2	
primary hyperoxaluria ii	2	



Conclusion

ORA

- Extracellular space / extracellular region
- Cell periphery / plasma membrane altered
- Pyruvate metabolism

GSEA

- Respiratory electron transport
- Protein synthesis / translation
- Collagen formation / ECM remodeling

Reduced mitochondrial activity

Reduced lipid oxidation

Obesity

Reference

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Thanks!